AI Shield

From image detection to evidence-aware verification

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A project story: what I built, what failed, what improved, and where the work stands.

ONE IMAGE

AI-generation detection

Recorded origin and editing evidence

Research evidence, followed by a working local service.

Can the prediction survive a change in source?

The goal expanded from a classifier to an evidence-aware verification tool.

01

Detect

Estimate whether an image shows visual evidence of AI generation.

02

Generalize

Test on new generators and image sources, not only familiar examples.

03

Explain evidence

Show available camera, editing and signed-history information.

An AI score is a prediction. Recorded history is separate evidence.

Six stages, each prompted by a limitation

Milestones show the development order without calendar dates.

1. **Face detector** — A working prototype, but a narrow image domain.
2. **Forensic branches** — Broader visual signals, reconstruction and face routing.
3. **Semantic + fusion** — Custom patch attention and a learned score combiner.
4. **Generalization audit** — A newer benchmark exposed the original model's weakness.
5. **Community Forensics** — A stronger detector, selected by controlled comparison.
6. **Provenance + service** — Repeatable inference and a combined local upload API.

The first detector worked, but its scope was narrow

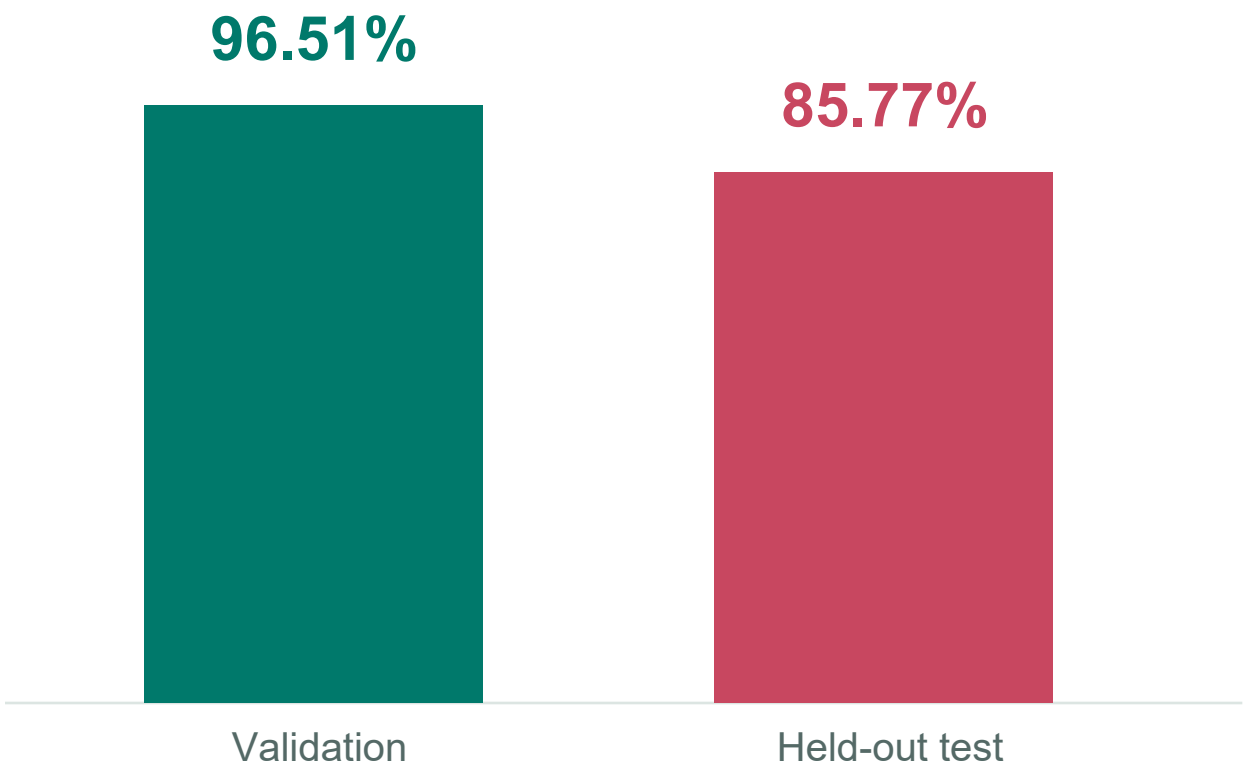
EfficientNetV2-B0 trained on OpenForensics face-manipulation images.

What I built

Training, evaluation and an image-upload demo for face manipulation.

What I learned

A strong validation result did not guarantee the same reliability on held-out images.



Accuracy; chart baseline = 0%

Built several forensic signals, then tested their value

Forensic = patterns in the pixels and image-processing traces.

EfficientNet-B0

General image classifier

Trained on balanced Tiny-GenImage; calibrated scores and saved branch outputs.

Real-only VAE

Reconstruction test

Built the encoder, decoder and training loop; tested whether unusual reconstruction errors flag AI.

FaceGate

Use the face model only when relevant

Checked face presence and routed the OpenForensics score only when a valid face was found.

Negative finding: VAE alone AUROC ~0.470; a naive weighted blend hurt performance.

Built a custom attention head over image patches

ViT = Vision Transformer; it represents an image as a sequence of patches.

Image patches → **Pretrained ViT backbone** → **Custom attention + refinement + pooling** → **Whole-image AI score**

856,578

trainable custom-head parameters

Backbone frozen; custom head trained. Implemented multi-head attention, refinement and attention pooling.

Visual modeling, not demonstrated caption checking or human-like reasoning.

Learned fusion improved the original benchmark

Original Tiny-GenImage fusion test: 1,400 images. These are historical results.



17 features

11 forensic + 6 semantic

94.07%

accuracy on this test

GLM = logistic regression: a learned score combiner.
AUROC measures ranking: 0.5 is chance; 1.0 is perfect. It is not accuracy.

Caught 54 additional generated images with only 1 additional false positive.

Newer images exposed a generalization problem

Re-ran the saved original AI Shield pipeline, without retraining it.

ORIGINAL BENCHMARK

0.9819

AUROC on the original
Tiny-GenImage fusion test

NEW DATASET v1

~0.606

AUROC on the new ordinary test
~0.599 on held-out GPT Image 2

Different test sets: this contrast shows the transfer problem, not a training curve.

Next: compare old AI Shield with Community Forensics (CF) on the same images.

Built Dataset v1 to test a harder question

Fixed image identities and splits; evaluated a generator excluded from our fine-tuning.

5,124

balanced images

2,562 real
2,562 generated



- **Training: 2,990**
- **Validation: 640**
- **Ordinary test: 640**
- **Unseen generator: 854**

Five generator families in development.
GPT Image 2 reserved for a separate test.

Held out from this project's training/validation; no claim about the pretrained model's full history.

Fine-tuned the detector in two controlled stages

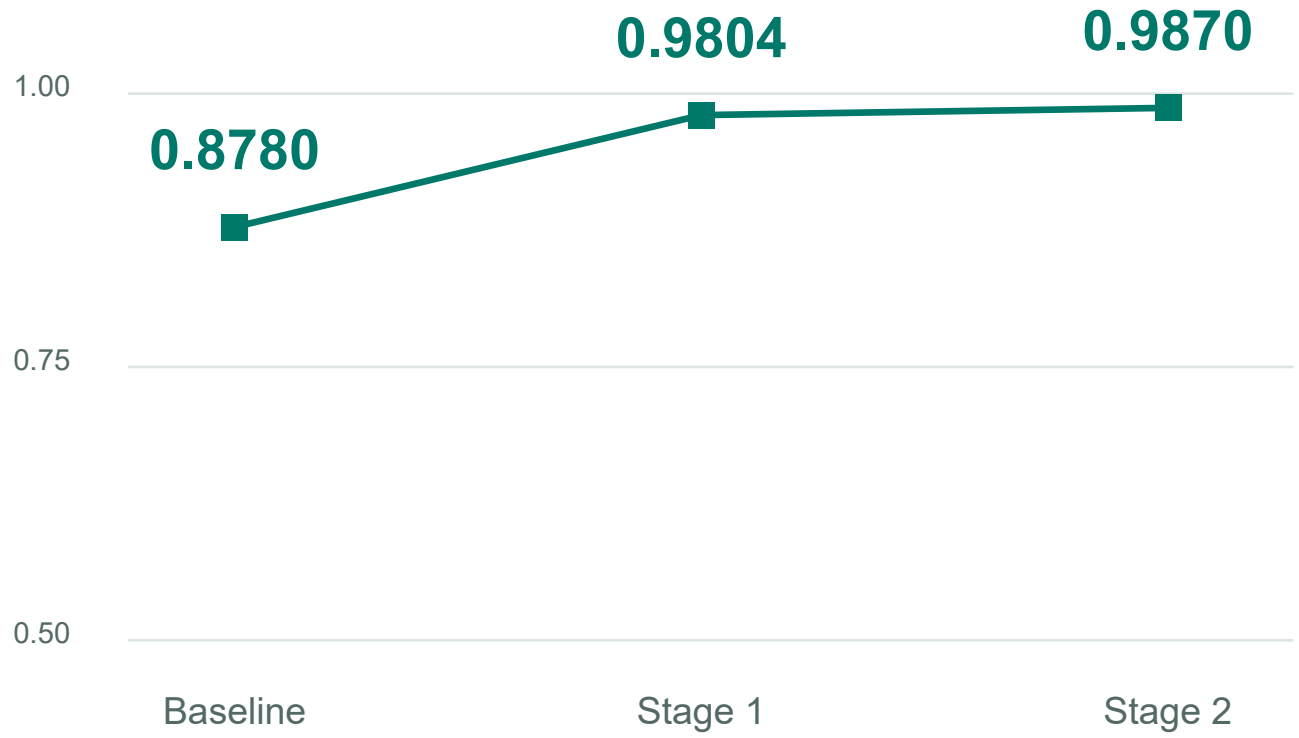
The checkpoint and decision threshold were selected on validation data.

Stage 1: classifier head

Three epochs; keep the pretrained visual backbone fixed.

Stage 2: final blocks

Three epochs; train the final two transformer blocks, normalization and head.



Validation AUROC improved from 0.8780 to 0.9870.

Fine-tuned CF became the strongest detector

Dataset v1 final evaluation. AUROC is ranking performance, not percentage accuracy.

MODEL	ORDINARY TEST AUROC	GPT IMAGE 2 AUROC
Old AI Shield fusion	0.6060	0.5990
Original CF	0.8860	0.9555
Fine-tuned CF	0.9884	0.9946

Fine-tuned accuracy: 95.16% ordinary test | 96.60% held-out-generator test.

Real-image false alarms: 20.63% -> 3.75% on the ordinary test.

Each CF model uses its validation-selected max-F1 cutoff; bar scales run from 0 to 1.

More models did not improve this detector

New fusion: fine-tuned CF + calibrated EfficientNet + FaceGate/OpenForensics.

MODEL	TEST AUROC	GPT IMAGE 2 AUROC
Fine-tuned CF alone	**0.9884**	**0.9946**
New fusion	**0.9854**	**0.9936**

The combination was slightly worse on both sets.
This is a separate experiment from the original 17-feature GLM.

Decision: select CF alone; pause further model-fusion experiments.

Locked the detector before building around it

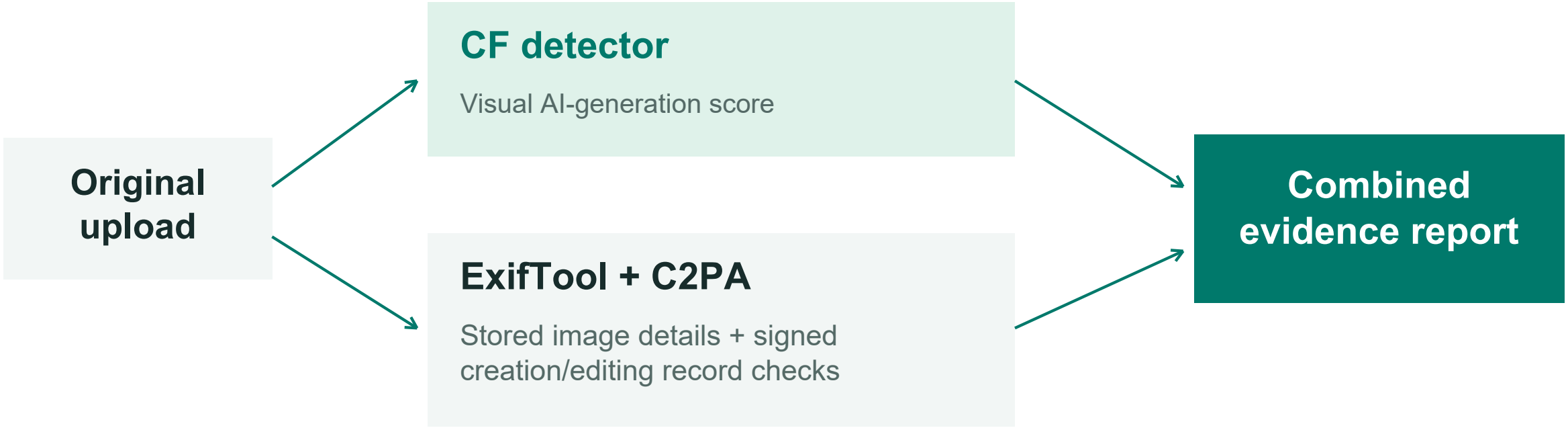
Frozen = same weights, same image preparation, same decision cutoff.

01	**Checkpoint hash**	A digital fingerprint confirms the exact fine-tuned model file.
02	**Exact preprocessing**	Orientation -> RGB -> resize -> crop -> normalization.
03	**Fixed threshold**	About 0.634, selected using validation F1 (precision/recall balance).
04	**Reference-image checks**	Six saved examples matched scores within tolerance and kept their labels.

These are implementation-consistency checks, not a new accuracy benchmark.

Added recorded provenance beside the AI score

FastAPI = the Python service that runs both checks for the future website.



38 tests passed | 6 reference images matched through the live upload API.

EXIF = stored camera/software/date details. C2PA = signed creation/editing records.

Missing metadata = unknown. Photoshop is not proof of AI. Credential validity is not scene truth.

Working locally; evaluation and deployment come next

The hosted Hugging Face demo still uses the older system.

WORKING NOW

Fine-tuned CF detector
Repeatable inference
EXIF / C2PA reporting
Combined local upload API

NEXT

JPEG / resize robustness
Artwork and edited-photo tests
Publish code + build website
Prepare public deployment

LATER

Image / claim consistency
Short-video support
Broader source evaluation

For discussion

Which evaluation would make the capstone most convincing?
How should uncertain or conflicting evidence be presented?